# 7. Area 100 KSA

## 7.1. Introduction

Area 100 KSA was introduced into the professional licence theoretical syllabus in response to the airlines' view that the student's pilot competencies, threat and error management, resilience and mental maths skills should be developed alongside and within theoretical knowledge subjects through the development and assessment of the attitude and skills required to apply this knowledge in cooperative and controlled scenarios.

Area 100 KSA is based on the ICAO Core Competencies that do not require the direct control of an aircraft, namely:

- Communication
- Leadership and teamwork
- Problem-solving and decision-making
- Situation awareness
- Workload management
- Application of knowledge

Additionally, students are required to demonstrate threat and error management across subjects, resilience, and a good knowledge of aviation related mental maths.

## 7.2. Importance and benefits of Area 100 KSA

The importance of a proper understanding of the need to implement Area 100 KSA and of its benefits is of paramount importance for ATOs.

Three main needs can be identified:

- allow students to apply the theoretical knowledge learnt during classroom lessons to effectively use threat and error management principles and make informed decisions;
- develop strong competencies with resilience to safely and efficiently manage both predictable and unpredictable situations;
- improve the students' understanding of what is being learnt during theoretical knowledge instruction and why.

A good application of Area 100 KSA results in students benefiting from:

- improved soft skills;
- improved attitude;
- improved motivation;
- better personal growth;
- improved problem solving abilities;
- greater big picture thinking;
- reinforced memory;
- strengthened mental maths calculation abilities.

## 7.3. Requirements for Area 100 KSA

In accordance with AMC3 ORA.ATO.230(a) of Regulation (EU) No 1178/2011, Area 100 KSA should comprise at least two summative assessments and one formative assessment covering the learning objectives of Subject Area 100 – Knowledge, Skills and Attitudes. Furthermore, written or oral mental maths tests should be organised with aviation related scenario-based mathematical problems. Further information can be found in AMC4 ORA.ATO.230(a) and subsequent GMs of Regulation (EU) No 1178/2011.

## 7.4. Learning process

The learning process is to be defined by considering inputs, defining development steps and assessing the outputs of the system.

### 7.4.1 Inputs of the learning process

To implement Area 100 KSA, ATOs should take into consideration the following input elements:

- the students' aptitudes as result of the selection process;
- the organisation of the training course with regard to the integration of theoretical instruction and practical flight training;
- the training tools used to deliver theoretical knowledge instruction;
- the training locations;
- the number of students;
- the required and the available resources.

A thorough analysis of these elements should be conducted with the aim of defining:

- the starting level of training to build competencies;
- the final level of training which depends on the level of knowledge, skills and attitude that trainees are expected to have at the end of their final summative assessment.

By determining the starting and the final levels, the learning process to build pilot competencies can be defined. The process should be spread across the entire theoretical knowledge instruction course and be progressive.

### 7.4.2 Steps to define the learning process

In defining the learning process, ATOs are recommended to follow some sequential steps.

1) Define which competencies are to be built at each stage of the training course and the expected level students should achieve.
2) Define how to develop competencies throughout the training course.

   This activity requires an adequate identification of the most suitable training tools for the scope of the planned activities. A non-exhaustive list of tools includes assignments, group exercises, written assessments, presentations and visits to aeronautical stakeholders.
3) Define how to observe the acquisition of competencies by defining the proper performance indicators for formative and summative assessments.
4) Define how to assess the competencies against the performance indicators for formative and summative assessments.

The observation and the assessment of competencies requires:

- a good and robust standardisation process of instructors and assessors;
- the preparation of an assessment plan that includes all the tasks that trainees are expected to execute during a specific assessment, their sequence, and the relation of the task sequence with the competencies to be observed;
- the analysis of the relevant threat and error management principles in the assessment of competencies.

Assessments should be designed so that they reflect the level of progress within the training course. The assessments may be different depending on the stage of the training course during which the assessment is conducted.

At the end of the final summative assessment, the student should have demonstrated a satisfactory level in all the required competencies.

### 7.4.3 Outputs of the learning process

At the end of process, ATOs should have robust and standardised data on the students' performance and experience during the course.

This set of data can be used for the purposes illustrated in *Section 4.2.5*.

## 7.5 Summative assessments

The content of a summative assessment is herewith illustrated and a practical example of a summative assessment is provided.

### 7.5.1 Content of a summative assessment

It is recommended that a summative assessment includes, as minimum, the following elements:

- the objective;
- the description of the assessment;
- the assessment of the observable competencies;
- the grading method;
- the feedback provided.

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| **Objective** | Summative assessments should collectively give students the opportunity to demonstrate competence in all the LOs in 100 02 and 100 03 of Area 100 KSA, as defined in AMC1 FCL.310; FCL.515(b); FCL.615(b); FCL.835(d).<br><br>A clear statement of the competencies to be evaluated during a summative assessment is expected.<br><br>An indication of the total length of the assessment is suggested. |
| **Description of the assessment** | The content and the elements of the assessment have to be described.<br><br>The competencies to be demonstrated are combination of knowledge, skills and attitude and not necessarily to be related to the direct operation of an aircraft. Different tools and aviation related scenario-based cases should be used. |

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| **Assessment of observable competencies** | The assessment should be objective, transparent and scenario-based.<br>The way competencies are assessed during the execution of the summative assessment should be clearly indicated to achieve an effective evaluation and to pinpoint potential performance indicators. To this aim, it is suggested to follow the guidance provided in GM2 ORA.ATO.230(a).<br>Assessors evaluating ICAO core competencies should be proficient in:<br>• Learning styles<br>• Teaching methods<br>• Facilitation techniques<br>• Threat and error management (TEM)<br>• The applicable competencies<br>• Content of the subjects<br>• Exercises to be delivered<br>Assessors should create an environment to trigger certain reflections and reactions from students in order to evaluate and provide feedback based on the competencies to be assessed. |
| **Grading** | Grading should be developed using table 2 of GM2.ORA.ATO.230(a) or equivalent in order to achieve effective means of measurement. If performance indicators not included in table 2 are to be used, such indicators have to be defined in an approved training manual.<br>Grading should only be conducted by assessors standardised to achieve inter-rater reliability and according to the specific ATO grading system. |
| **Feedback** | A debriefing with the students should be provided as shortly after the assessment as possible, to ensure that each student receives proper feedback on his or her performance and areas of improvement.<br>During the debriefing, the assessor should act as a facilitator and let the students initially provide their own feedback. The feedback from the assessor should emphasise areas of and opportunities for improvement, preferably with actionable suggestions on how to improve, as well as areas of excellence.<br>Feedback should, whenever possible, be aimed to motivate the students to improve and enlighten the students with where to improve, rather than informing a student of what they are not doing right. Observed examples may, of course, be useful in illustrating the areas of improvement, but should come with suggestions for improvement.<br>Providing feedback should be standardised, as any part of the assessment. For the summative assessment the feedback should also be recorded and added to the student's training records.<br>With regard to standardisation for feedback, it is important for the CTKI to instil in the assessors the aim of the subject and the aim of each individual assessment to ensure that feedback, areas of emphasis and the expected level of students, all align with the aim of the particular assessment and the training organisation as a whole. For each assessment the areas to be covered in the assessment and subsequent feedback, should be defined.<br>Feedback of students may prove to be very useful in evaluating and improving the assessments, the subject and the training course, but should always be de-identified before being used. |

### 7.5.2 Practical example of a summative assessment

**Objective**

This summative assessment consists of a scenario-based exercise to be performed in teamwork simulating real-world aviation-based situations. The exercise permits to evaluate the following students' competencies:

- Application of knowledge
- Leadership and teamwork
- Workload management
- Communications
- Problem-solving and decision-making
- Situation awareness

Additionally, UPRT elements and resilience are evaluated.

The assessment lasts about 1 hour.

**Description of the assessment**

In this example, a working group composed of two students represents the flight crew of a commercial aircraft. The composition of the group is defined by the ATO and, in case the number of students and assessors permits, two working groups can be assessed at the same time.

During the assessment one student simulates the role of pilot flying (PF) and the other one the role of pilot monitoring (PM). Roles are switched during the exercise.

The exercise includes one main activity to be done in teamwork (e.g. building with bricks, solving a puzzle), with the addition of the following activities to be carried out in parallel to the main one once an input from the assessor is received.

1) ATC standard communications addressed to the crew

   The assessor acts as ATCO simulating a real operative environment and instructs the crew requesting to perform a sequence of actions.

   The PM replies to the relevant communications. The PF has autopilot mode cards with selectable values representing simplified autopilot functions and manages the aircraft autopilot by choosing the correct mode cards and values to comply with the required ATC instructions.

2) Autopilot management

   The PF passes the mode card confirming the action to perform. The PM checks the PF input correcting him or her if necessary.

3) UPRT situations using pictures representing the PFD

   The PF recognises the situation declaring the recovery actions to perform. The PM checks the PF decisions correcting him or her if necessary.

4) Unusual or emergency situations

   The assessors present to the flight crew real-world situations using relevant pictures of flight displays. After an evaluation of relevant information, the students consider the available options and make a shared decision. The actions are communicated by the PM to the ATCO and performed by the PF who selects the correct autopilot mode cards and values.